COLLEGE OF ENGINEERING & TECHNOLOGY

Department: Basic and Applied Sciences

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Course: (EBA0201)

Sheet 2

Calculate the derivative of the following function:

- $\frac{d}{dx}(x^n) = n(x)^{n-1}.\ x'$

1. $y = x^2 + 3$
2. $s = 5t^3 - 3t^5$
3. $y = \frac{4x^3}{3} - x$
4. $y = 6x^2 - 10x - 5x^{-2}$
5. $y = (3x - 5)^3$
6. $y = (x^4 + 3x)^{-1}$
7. $y = \frac{(3x^2 - \pi x)^4}{6}$

- $\frac{d}{dx}(\sqrt{x}) = \frac{1}{2\sqrt{x}}.\ x'$

1. $y = (\sqrt{x} - 1)^2 - (x^2 + 1)^4$

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2. $y=\sqrt{2x}+\sqrt{5x}$

3. $y=\sqrt{x^5}-\sqrt[4]{x^9}$

- $\frac{d}{dx}(u(x)v(x))=u(x)v'(x)+v(x)u'(x)$
- $\left[\frac{u(x)}{v(x)}\right]'=\frac{u'(x)v(x)-u(x)v'(x)}{v^2(x)}$

1. $y=x^2(x^4+2x)^2$

2. $y=x(x^3+3x^2)^{-1}$

3. $y=x^4(x-5)^6$

4. $y=(x^2+3x)(x^2+2x)^2$

5. $y=\frac{2x+1}{x}$

6. $y=\frac{2x^4+3x^2-1}{x^2}$

7. $y=\frac{x^2-3x+2}{x+3}$

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- $\frac{d}{dx}\left[e^{x}\right]=e^{x}.x'$

1. $y=2e^{x}$
2. $y=2e^{-x}$
3. $y=e^{x+2}$
4. $y=\sqrt{2x}+(x+2)^{2}+\frac{2+x}{x}+2e^{2x}$

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